Transformation and deformation of Sea Waves

- As the waves move from deep water towards shallow water, they are transformed due to:
- Water depth change
- Variation in sea bottom configuration
- Existence of obstacles (islands, breakwater)



Transformation and deformation of Sea Waves

As a result:

- Wave height
- Wave direction may change
- Wave period?

$$C = \frac{gT}{2\pi}\tanh(\frac{2\pi d}{L})$$

Water waves travel slower in shallower water.

$$L = \frac{gT^2}{2\pi}\tanh(\frac{2\pi d}{L})$$

The slower the speed of the wave the shorter its wavelength will be.

This keeps the frequency the same.

Transformation and deformation of Sea Waves

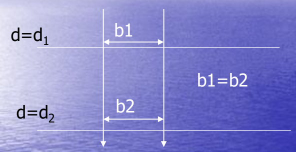
- Wave shoaling (dalga sığlaşması)
- Wave refraction (dalga kırınımı, sapması)
- Wave diffraction (dalga sapması, dönmesi)
- Wave breaking (dalga kırılması)

Wave shoaling (dalga sığlaşması)

Wave height change due to varying depth

WAVE SHOALING

- If the crest line is parallel to the bottom contours, b1=b2



Energy flux is conserved betw wave rays $=EC_g b=constant$
$C_g=Cn$

$E_1C_1n_1b_1=E_2C_2n_2b_2$

$E_2/E_1=(C_1n_1b_1)/(C_2n_2b_2)$

WAVE SHOALING

$E_2/E_1=(C_1n_1b_1)/(C_2n_2b_2)$

$$\bar{E}=\frac{1}{8}\gamma H^2$$

$$\frac{H_2}{H_1}=\sqrt{\frac{n_1}{n_2}\frac{C_1}{C_2}}$$

$$C=\frac{gT}{2\pi}\tanh(\frac{2\pi d}{L})$$

$$n=\frac{1}{2}(1+\frac{2kd}{Sinh(2kd)})$$

İf H_1, n_1, C_1 , b_1 belongs to deep water

$$\frac{H}{H_0}=\sqrt{\frac{n_0}{n}\frac{C_0}{C}}=\sqrt{\frac{1}{2}\frac{1}{n}\frac{1}{\tanh(kd)}}=K_s$$

Wave shoaling (dalga sığlaşması)

- Example:
- H_1=4m T=8 s at d_1=200m
- Find H_2 at d_2=20m

$L_0=1.56T^2=100$m
$d_1/L_0=200/100=2>0.5$ then DW
$d_2/L_0=20/100=0.2<0.5$ then IW

Wave shoaling (dalga sığlaşması)

$$\frac{H}{H_0}=\sqrt{\frac{n_0}{n}\frac{C_0}{C}}=\sqrt{\frac{1}{2}\frac{1}{n}\frac{1}{\tanh(kd)}}=\frac{H}{H_0}=K_s$$

At d=20m, $d/L_0=0.2$ from Wave Table tanh(kd)=0.888
n=0.5(1+0.335)=0.6675

Ks=0.918
H/4=0.918
H=3.67m

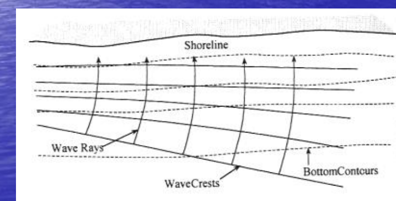
WAVE SHOALING

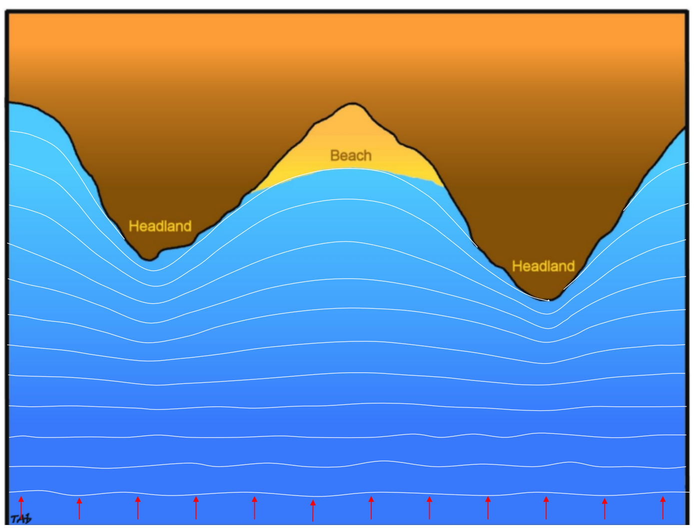
GRAVITY WAVE TABLE

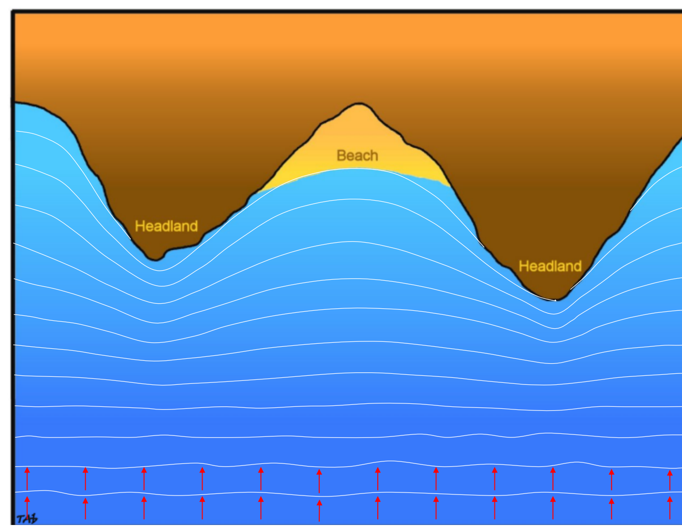
$\frac{d}{L_0}$	$\tanh\frac{2\pi d}{L}$	$\frac{d}{L}$	$\frac{2\pi d}{L}$	$\sinh\frac{2\pi d}{L}$	$\cosh\frac{2\pi d}{L}$	$\frac{4\pi d/L}{\sinh 4\pi d/L}$	K_s
0,00	0,000	0,0000	0,000	0,000	1,00	1,000	-
01	248	0403	253	256	03	0,958	1,44
02	347	0576	362	370	07	918	23
03	420	0714	448	463	10	877	12
04	480	0833	523	548	14	839	06
0,05	0,531	0,0942	0,592	0,627	1,18	0,800	1,02
06	575	104	655	703	22	763	0,993
07	614	114	716	778	27	725	971
08	649	123	774	854	31	690	955
09	681	132	831	0,930	36	654	942
0,10	0,709	0,141	0,886	1,01	1,42	0,621	0,933
11	735	150	940	08	48	587	926
12	759	158	0,994	16	54	555	920
13	780	166	1,05	25	60	524	917
14	800	175	10	33	67	494	915
0,15	0,818	0,183	1,15	1,42	1,74	0,465	0,913
16	835	192	20	52	82	437	913
17	850	200	26	61	90	410	913
18	864	208	31	72	1,99	384	914
19	877	217	36	82	2,08	359	916
0,20	0.888	0.225	1,41	1,94	2,18	0,335	0,918

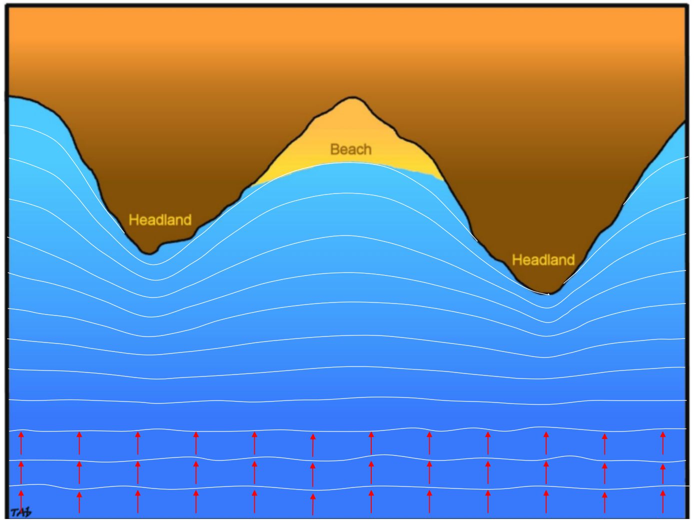
WAVE REFRACTION

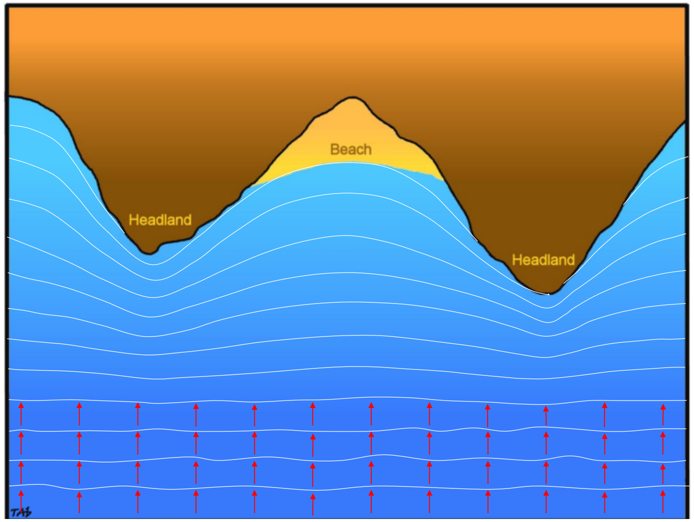
- When waves approach the shore at an angle= refraction
- During refraction waves bend to align themselves with the bottom contours
- Wave direction becomes more perpendicular to the shore








Beach
Headland
Headland
TAS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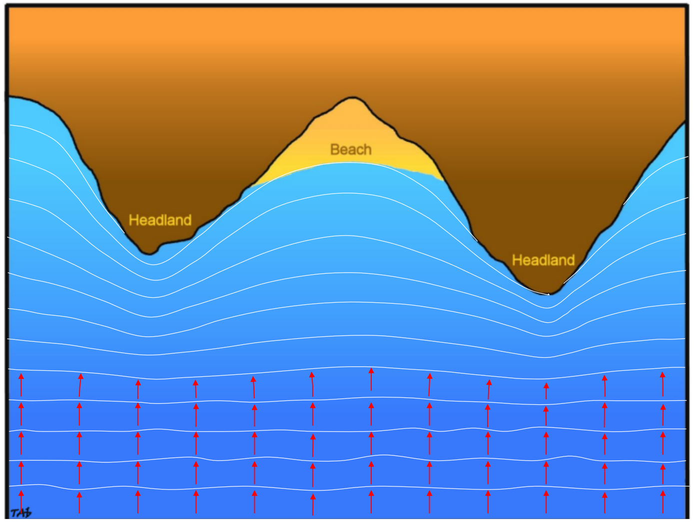
TAS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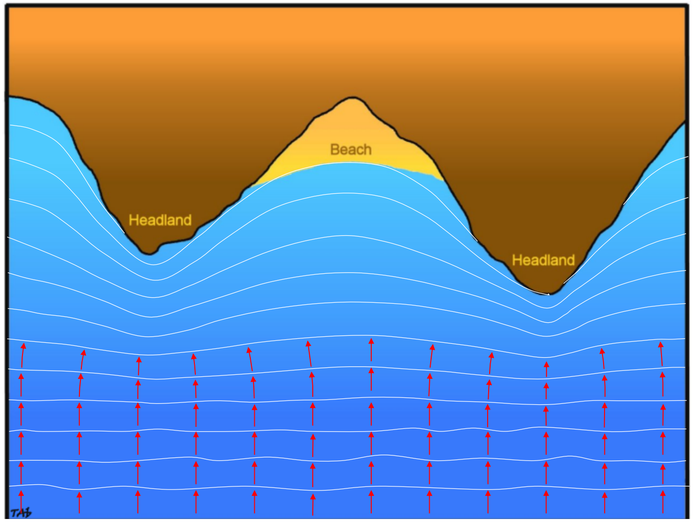
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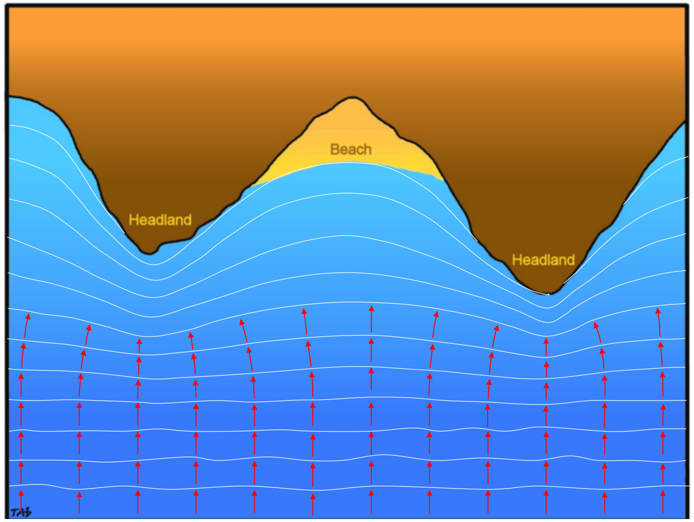
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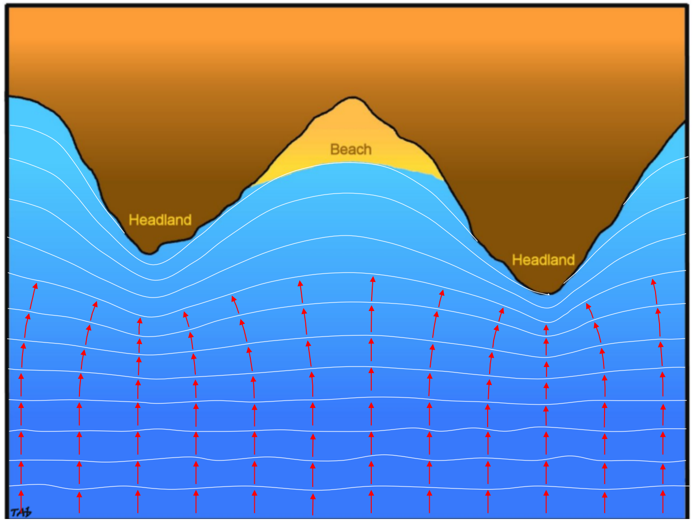
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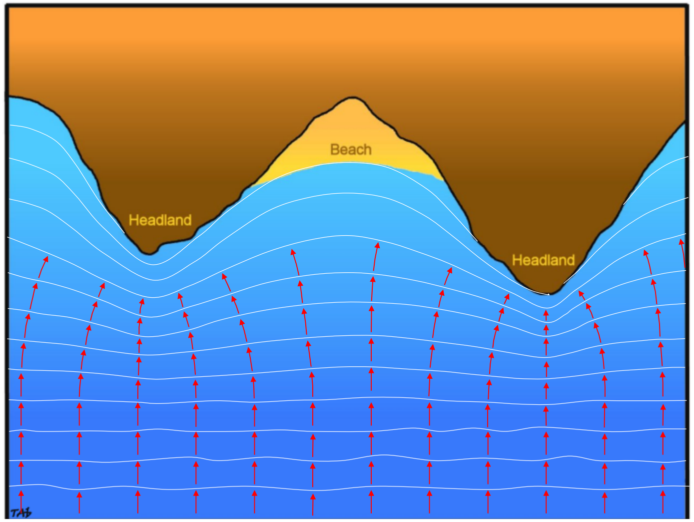
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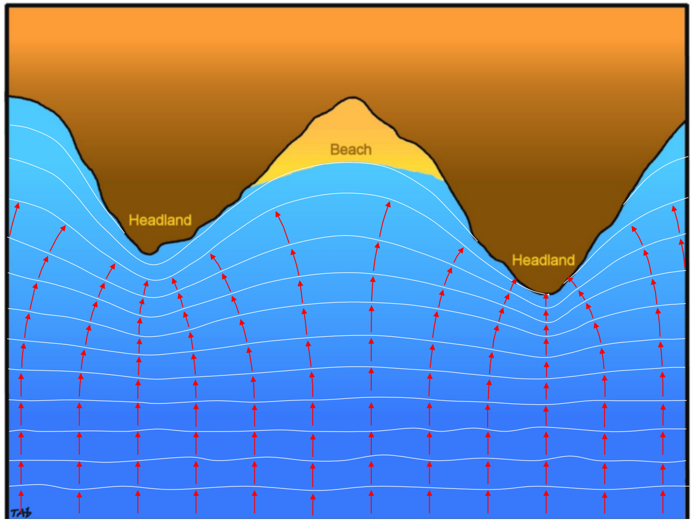
Beach
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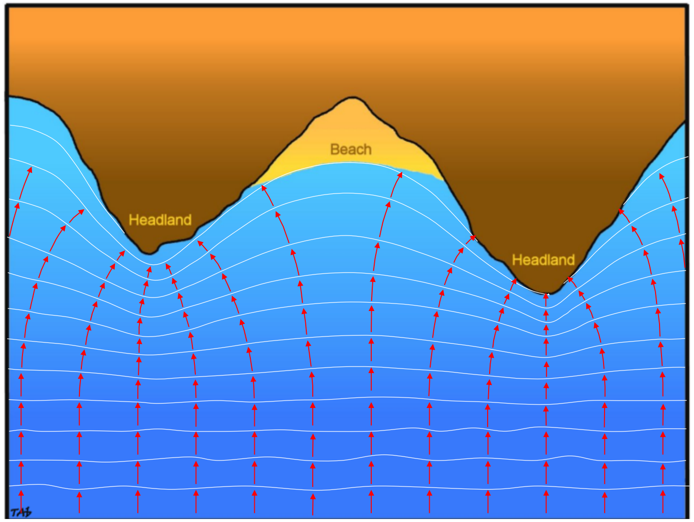
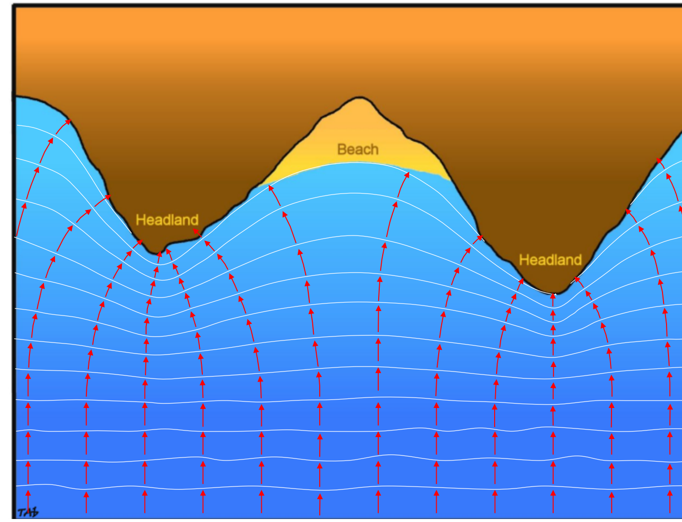
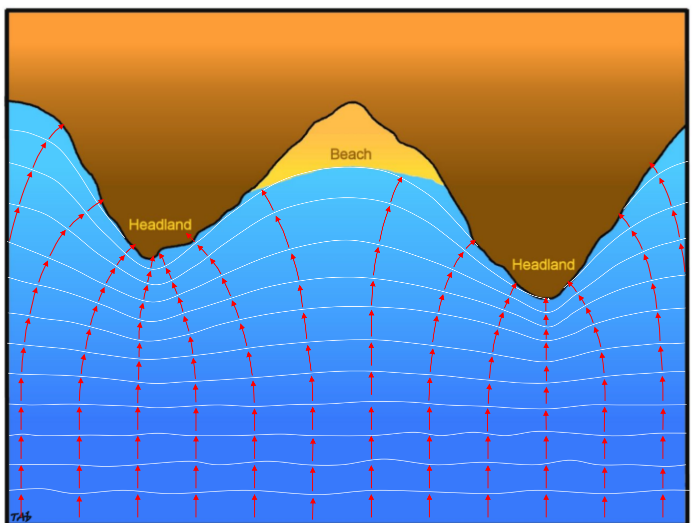
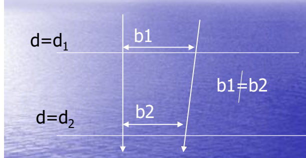
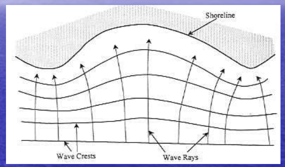






WAVE SHOALING+ REFRACTION





Energy flux is conserved betw wave rays =ECgb=constant

$E_1C_1n_1b_1=E_2C_2n_2b_2$

$E_2/E_1=(C_1n_1b_1)/(C_2n_2b_2)$

WAVE SHOALING+ REFRACTION

$E_2/E_1=(C_1n_1b_1)/(C_2n_2b_2)$

$$\bar{E}=\frac{1}{8}\gamma H^2$$

$$\frac{H_2}{H_1}=\sqrt{\frac{n_1}{n_2}\frac{C_1}{C_2}}\sqrt{\frac{b_1}{b_2}}$$

$$C=\frac{gT}{2\pi}\tanh(\frac{2\pi d}{L})$$

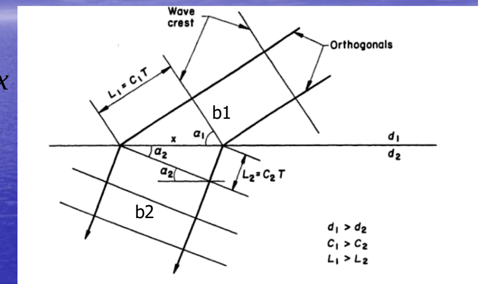
$$n=\frac{1}{2}(1+\frac{2kd}{\mathrm{Sinh}(2kd)})$$

İf H_1, n_1, C_1 , b_1 belongs to deep water

$$\frac{H}{H_0}=\sqrt{\frac{n_0}{n}\frac{C_0}{C}}\sqrt{\frac{b_0}{b}}=\sqrt{\frac{1}{2}\frac{1}{n}\frac{1}{\tanh(kd)}}\sqrt{\frac{b_0}{b}}=K_sK_r$$

Wave refraction (dalga kırınımı)

$$\frac{b_1}{\cos\alpha_1}=\frac{b_2}{\cos\alpha_2}=x$$



α is the angle between wave ray and the orthogonal of bottom contour line

Wave refraction (dalga kırınımı)

$$\frac{b_1}{\cos\alpha_1}=\frac{b_2}{\cos\alpha_2}$$

$$\sqrt{\frac{b_1}{b_2}}=\sqrt{\frac{\cos\alpha_1}{\cos\alpha_2}}$$

$$K_r=\sqrt{\frac{b_0}{b}}=\sqrt{\frac{\cos\alpha_0}{\cos\alpha}}$$

Wave refraction (dalga kırınımı)

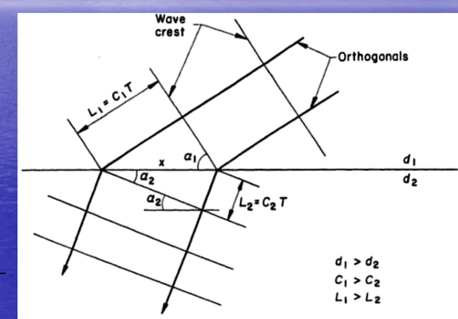
How will we find α?

$$\sin\alpha_1=\frac{C_1T}{x}=\frac{L_1}{x}$$

$$\sin\alpha_2=\frac{C_2T}{x}=\frac{L_2}{x}$$

SNELL'S LAW

$$\frac{\sin\alpha_1}{\sin\alpha_2}=\frac{C_1}{C_2}=\frac{L_1}{L_2}$$



Wave refraction (dalga kırınımı)

How will we find α?

Snell's law of wave refraction:

$$\frac{\sin\alpha}{\sin\alpha_0} = \frac{C}{C_0} = \tanh\frac{2\pi d}{L}$$

WAVE SHOALING+ REFRACTION

Example:
H_{s0}=3m, T_s=8 sec, α_0=30^0

Hs,α=? At d=10m

WAVE SHOALING+ REFRACTION

Example:
H_{s0}=3m, T_s=8 sec, α_0=30^0

Hs,α=? At d=10m

d=10m, from wave table:
L_0=100m, d/L_0=0.1, d/L=0.14, tanh(kd)=0.71, n=0.81

$$K_s = \sqrt{\frac{1}{2}\frac{1}{n}\frac{1}{\tanh(kd)}} = 0.93$$

$$\frac{\sin\alpha}{\sin\alpha_0} = \frac{C}{C_0} = \tanh\frac{2\pi d}{L}$$

$$K_r = \sqrt{\frac{\cos\alpha_0}{\cos\alpha}}$$

α=20.9^0

$$K_r = \sqrt{\frac{\cos 30}{\cos 20.9}} = 0.96$$

Hs=3x0.93x0.96
=2.7m

WAVE BREAKING (DALGA KIRILMASI)

- Wave shoaling causes wave height increase to infinity in very shallow area
- However, there is a physical limit the steepness of the waves (H/L)
- When this limit is exceeded, the waves break and dissipates its energy

WAVE BREAKING (DALGA KIRILMASI)

GRAVITY WAVE TABLE

$\frac{d}{L_0}$	$\tanh\frac{2\pi d}{L}$	$\frac{d}{L}$	$\frac{2\pi d}{L}$	$\sinh\frac{2\pi d}{L}$	$\cosh\frac{2\pi d}{L}$	$\frac{4\pi d/L}{\sinh 4\pi d/L}$	K_s
0,00	0,000	0,0000	0,000	0,000	1,00	1,000	-
01	248	0403	253	256	03	0,958	1,44
02	347	0576	362	370	07	918	23
03	420	0714	448	463	10	877	12
04	480	0833	523	548	14	839	06
0,05	0,531	0,0942	0,592	0,627	1,18	0,800	1,02
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08	649	123	774	854	31	690	955
09	681	132	831	0,930	36	654	942
0,10	0,709	0,141	0,886	1,01	1,42	0,621	0,933
11	735	150	940	08	48	587	926
12	759	158	0,994	16	54	555	920
13	780	166	1,05	25	60	524	917
14	800	175	10	33	67	494	915
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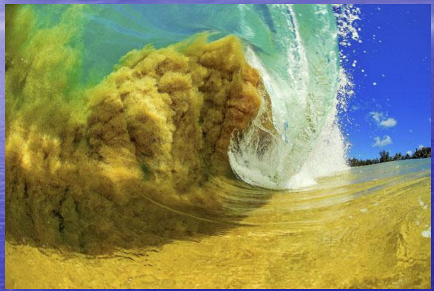
16	835	192	20	52	82	437	913
17	850	200	26	61	90	410	913
18	864	208	31	72	1,99	384	914
19	877	217	36	82	2,08	359	916
0,20	0.888	0.225	1,41	1,94	2,18	0,335	0,918



Wave Breaking Criteria

Deep water: $\frac{H_o}{L_o} = 0.142 = \frac{1}{7}$ (Steepness criteria)

Intermediate water: $\left(\frac{H}{L}\right)_{max} = 0.142 \tanh\left(\frac{2\pi d}{L}\right)$

Shallow water: $\frac{H_b}{d_b} = 0.78$ (Spilling breaker criteria)

Types of Wave Breaking

a) Spilling breaking wave

b) Plunging breaking wave

c) Surging breaking wave

d) Collapsing breaking wave

Breaker Types

Spilling	$\xi_o < 0.5$
Plunging	$0.5 < \xi_o < 3.3$
Surging/collapsing	$\xi_o > 3.3$

$$\xi_o = \frac{\tan\theta}{\sqrt{H_o / L_o}}$$

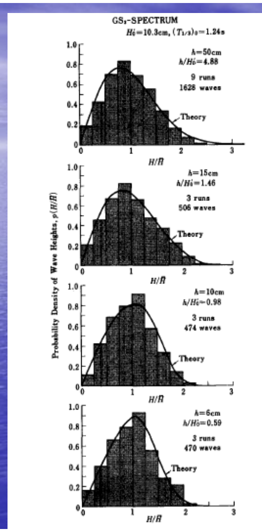
Surf similarity parameter / Irribarren number

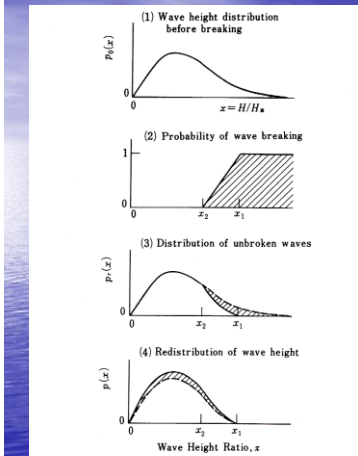
WAVE BREAKING (DALGA KIRILMASI)

- Engineering interests:
- Do waves break at the construction depth?
- What is the wave height Hs and Hmax at the construction depth if breaking occurs

Computational Model of Random Wave Breaking

- Wide spatial spread in the region of breaking
- Mechanism of wave attenuation within the surf zone is very difficult to understand because of complicated nature of the wakes, turbulance and air entrainment
- But it is possible to analyze the change in the wave height



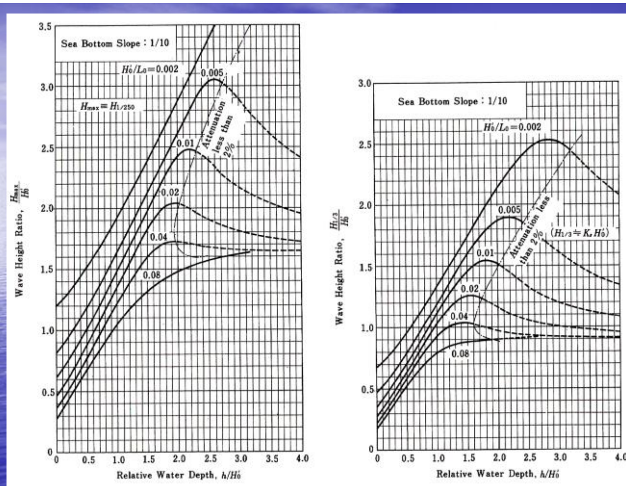


Explanatory sketch of the model of random waves breaking.

$$\frac{H_b}{L_0} = A\left\{1 - \exp\left[-1.5\frac{\pi h}{L_0}(1 + 15\tan^{4/3}\theta)\right]\right\}$$

Breaker index

A=0.18-0.12



Diagrams for the estimation of wave heights in the surf zone (sea bottom slope of 1/10).

•Goda,Y.2000

$$H_{1/3} = \begin{cases} K_s H'_0 & d/L_0 \geq 0.2 \\ \min\{(\beta_0 H'_0 + \beta_1 h), \beta_{maks} H'_0, K_s H'_0\} & d/L_0 < 0.2 \end{cases}$$

$$\beta_0 = 0.028 (H'_0/L_0)^{-0.38} \exp\left[20(\tan\theta)^{1.5}\right]$$

$$\beta_1 = 0.52 \exp[4.2 \tan\theta]$$

$$\beta_{maks} = \text{maks}\{0.92, 0.32 (H'_0/L_0)^{-0.29} \exp[2.4 \tan\theta]\}$$

$$H_{maks} = \begin{cases} 1.8 K_s H'_0 & d/L_0 \geq 0.2 \\ \min\{(\beta_0^* H'_0 + \beta_1^* d), \beta_{maks}^* H'_0, 1.8 K_s H'_0\} & d/L_0 < 0.2 \end{cases}$$

$$\beta_0^* = 0.052 (H'_0/L_0)^{-0.38} \exp\left[20(\tan\theta)^{1.5}\right]$$

$$\beta_1^* = 0.63 \exp[3.8 \tan\theta]$$

$$\beta_{max}^* = \text{maks}\left\{1.65,\ 0.53 \left(\frac{H'_0}{L_0}\right)^{-0.29} \exp[2.4 \tan\theta]\right\}$$

Wave height estimation

in the surf zone

Example 3.7

Estimate the significant wave height at the depth of $h = 8$ m by means of Eq. (3.25), when swell with $H_0' = 6$ m and $T_{1/3} = 15$ s is attacking a coast with a slope of 1/50.

REGULAR WAVE CONCEPT

3 Cases for breaking

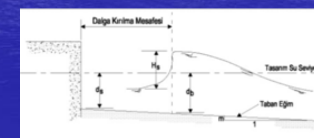
ds>db	nonbreaking
d'<ds<db	breaking
ds<d'	broken

ds=structure construction depth

Breaker travel distance

Xp=(4.0-9.25m)Hb

d'= db-Xp m



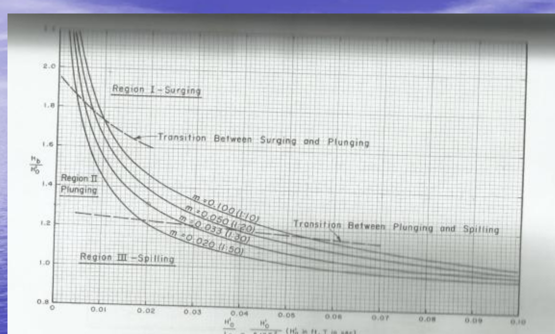
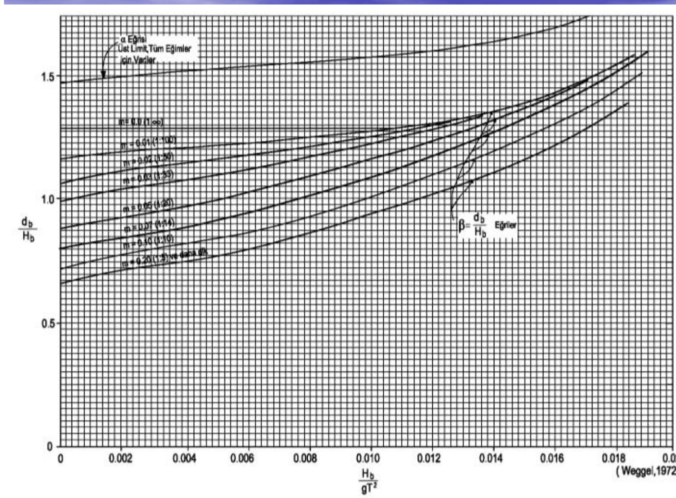
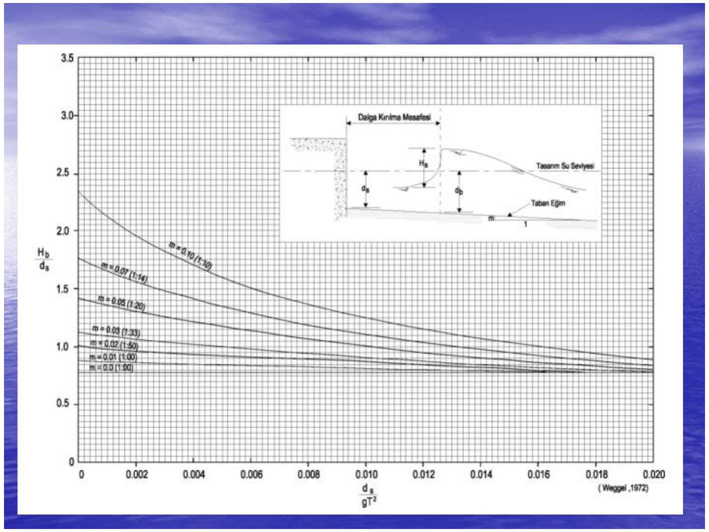


Figure 9.6 Breaker Height index versus deepwater wave steepness




Dalga Kırılma Mesafesi
Tasarım Su Seviyesi
Taban Eğim
m = 0.10 (1:10)
m = 0.07 (1:14)
m = 0.05 (1:20)
m = 0.03 (1:33)
m = 0.02 (1:50)
m = 0.01 (1:00)
m = 0.0 (1:00)
Hb/ds
ds/gT²
(Weggel ,1972)